

Tutorial 2

Sets and Systems of linear equations

1. Let $\mathcal{U}$, $\mathcal{L}$ and $\mathcal{D}$ respectively be the set of all upper triangular, all lower triangular and all diagonal 3×3 matrices, and let I be the 3×3 identity matrix. For each of the following statements say whether it is true or false, and give a reason for your answer.

(a) $I \in \mathcal{D}$

Solution: True

(b) $\{I\} \in \mathcal{D}$

Solution: False.

$\{I\}$ is a **set** (that contains only the identity matrix I), and the elements of $\mathcal{D}$ are **matrices**, not sets of matrices.

(c) $\{I\} \subseteq \mathcal{U}$

Solution: True. On the left-hand side, we have a set that contains only the 3×3 identity matrix I . On the right-hand side, we have a set that contains all upper triangular matrices.

The 3×3 identity matrix I is an upper triangular matrix. Hence the set on the left-hand side is a subset of the set on the right-hand side.

(d) $\emptyset \subseteq \mathcal{D}$

Solution: True. The empty set is subset of any set.

(e) $\mathcal{D} \subseteq \mathcal{U} \cap \mathcal{L}$

Solution: True

(f) $\{A^T : A \in \mathcal{U}\} \subseteq \mathcal{U}$

Solution: False. $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \in \mathcal{U}$, but $A^T = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix} \notin \mathcal{U}$, so $\{A^T : A \in \mathcal{U}\}$ is not a subset of $\mathcal{U}$.

(g) $\{A^T : A \in \mathcal{D}\} = \mathcal{D}$

Solution: True (use the fact that $A^T = A$ for every $A \in \mathcal{D}$)

(h) $\{A^T : A \in \mathcal{U}\} = \mathcal{L}$

Solution: True (a matrix is upper triangular if and only if its transpose is lower triangular). Proof: A^T is lower triangular if and only if $(A^T)_{ij} = A_{ji} = 0$ whenever $i < j$, which is true if and only if A is upper triangular.

2. Decide whether the following statement is true or false:

“For any two sets A and B , at least one of the following statements is true:

(a) $A \subseteq B$ (b) $B \subseteq A$ (c) $A = B$.”

If the statement is true, prove it. If it is false, give a counter-example.

Solution: The statement is false. A possible counter-example: Let $A = \{0, 1\}$, $B = \{1, 2\}$. Then A is not a subset of B , B is not a subset of A and $A \neq B$.

3. Determine which of the following statements are true:

(a) The empty set is a subset of every set.

Solution: True

(b) If $A \subseteq B$, then $A = B$.

Solution: False

(c) If $A = B$, then $A \subseteq B$.

Solution: True

(d) Since $\emptyset$ is a member of $\{\emptyset\}$, $\emptyset = \{\emptyset\}$.

Solution: False. $\emptyset$ contains no elements while $\{\emptyset\}$ contains exactly one element (the empty set).

(e) The set of all symmetric 3×3 matrices and the set of all skew-symmetric 3×3 matrices are disjoint.

Solution: False. Matrix $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is both symmetric and skew symmetric.

4. Let A be a 3×3 matrix **in row echelon form**. Consider the equation

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

- (a) Is it possible for this equation to have:

(i) A unique solution (ii) No solution (iii) Infinitely many solutions?

Give an example of A in each possible case. If you think some case(s) cannot occur, explain why not.

Solution:

(i) Yes. Any matrix A in row echelon form which has all the diagonal entries non-zero will give a unique solution. (To see why this is the case, remember that once A is in row-echelon form, we use backsubstitution to find the solution.) Think about it, what is this unique solution?

(ii) No. $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ is always a solution, regardless of what A is.

(iii) Yes, For example if $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 0 \end{pmatrix}$. Any matrix A with fewer than 3 pivots will be an example.

- (b) Find a condition, in terms of only the diagonal entries (the a_{ii}) of A , that distinguishes between the cases (i) and (iii) above.

Solution: Case (i): $a_{ii} \neq 0$ for all i .

Case (iii): At least one $a_{ii} = 0$. (This will be the case if and only if $a_{33} = 0$. Why?)

5. Four augmented matrices are given below with their associated variables. Are any of these augmented matrices **not** in row-echelon form? If so, which one(s)?

For the ones that are in row-echelon form:

- (i) say which variables are free and which are dependent;
(ii) solve the system associated with the augmented matrix and write down the **solution set**. Use the **vector form** of the solution to write down the solution set (i.e. write the solution set as a set of vectors).

(a) $\left(\begin{array}{ccc|c} 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 1 \end{array} \right) \quad x, y, z$

Solution:

(i) x is free, y, z are dependent

(ii) The solution is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, t \in \mathbb{R}$. The **solution set**

(which is what was asked for) is $\left\{ \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} : t \in \mathbb{R} \right\}$.

Make sure that you know how to write the solution as a *set*.

(b) $\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{array} \right) \quad x, y, z$

Solution: The matrix is not in row-echelon form.

(c) $\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 8 \\ 0 & 0 & 1 & 1 & 1 & 8 \\ 0 & 0 & 0 & 5 & 0 & 0 \end{array} \right) \quad x, y, z, w, v$

Solution:

(i) y, v are free, x, z, w are dependent

(ii) Solution set is $\left\{ \begin{pmatrix} 0 \\ 0 \\ 8 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} : s, t \in \mathbb{R} \right\}$

(d) $\left(\begin{array}{ccccc|c} 4 & -1 & 3 & 7 & 1 & 2 \\ 0 & 0 & 3 & 4 & 9 & 1 \\ 0 & 0 & 0 & 0 & 0 & 5 \end{array} \right) \quad x_1, x_2, x_3, x_4, x_5$

Solution: The solution set is $\emptyset$ (this system has no solutions).

6. Sketch the following set:

$$\{(x, n) \in \mathbb{R} \times \mathbb{N} : 1 \leq x < 5 \text{ and } 0 \leq n \leq x\}.$$

Solution: Since the set is a subset of $\mathbb{R} \times \mathbb{N}$, the second “coordinate” must be a natural number. The left endpoints of all the line segments are included, the right endpoints excluded.

